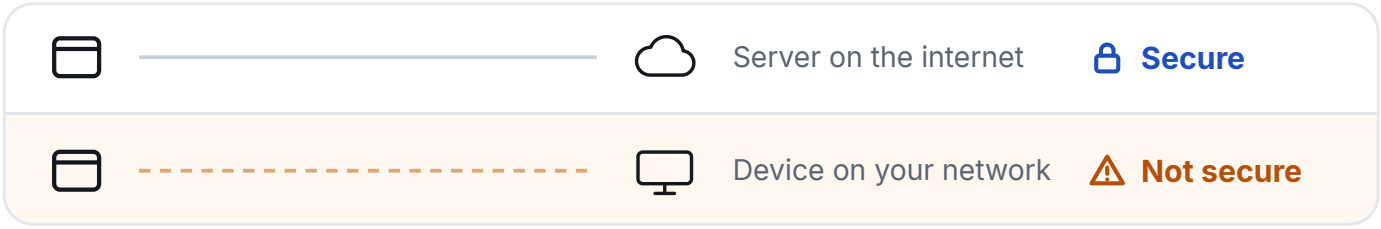
DRAFT PROPOSAL

Your browser trusts a server across the world. Not the TV next to you.



HOW IT WORKS

1 Pair once `482 913`

The user confirms a PIN or QR code. Devices authenticate each other, offline, without a certificate authority.

2 Browser keeps trust

In a local trust store, shared across sites but never exposed to them.

3 Sites get a pipe

To the peer the user picked, for that site only: data channels or WebTransport.

BUILDING BLOCKS

Trust between devices

Trust ceremonies, a local trust store and per-site grants.

Local data pipes

RTCDataChannel-style channels, or WebTransport streams and datagrams.

```
await lp2p.connect()
```

Local HTTPS draft

The browser as a client of HTTPS servers on the LAN, validated against the same trust.

SECURITY MODEL

- **Pages never see the network.** No addresses, no discovery results: the browser shows the picker.
- **Same-origin by default.** Other origins need an explicit allowedOrigins opt-in, as with CORS.
- **The remote origin is asserted, not proven.** Like the Origin header, it routes and isolates but never grants anything.
- **Nothing about a site before authentication.** The origin only reaches a verified peer.
- **Authentication, not proximity.** Being on the same network grants nothing; safe on café Wi-Fi.
- **Trust is visible and revocable.** Removing a device or grant closes its sessions.

Built for offline collaboration, file sharing, local multiplayer, smart home and device setup, media and casting, home labs, classrooms.

Spec, explainer, threat model and questionnaire

A Web platform API proposal in the WICG.

`github.com/WICG/local-peer-to-peer`